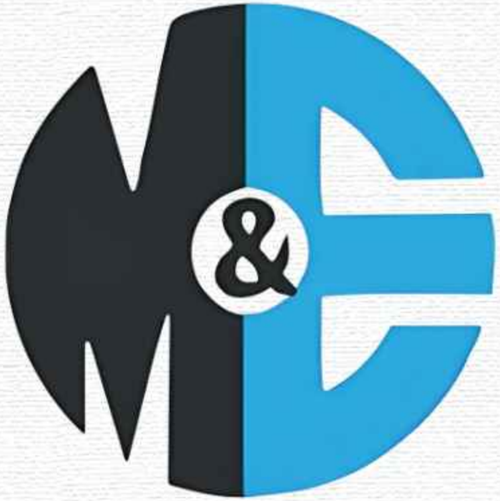
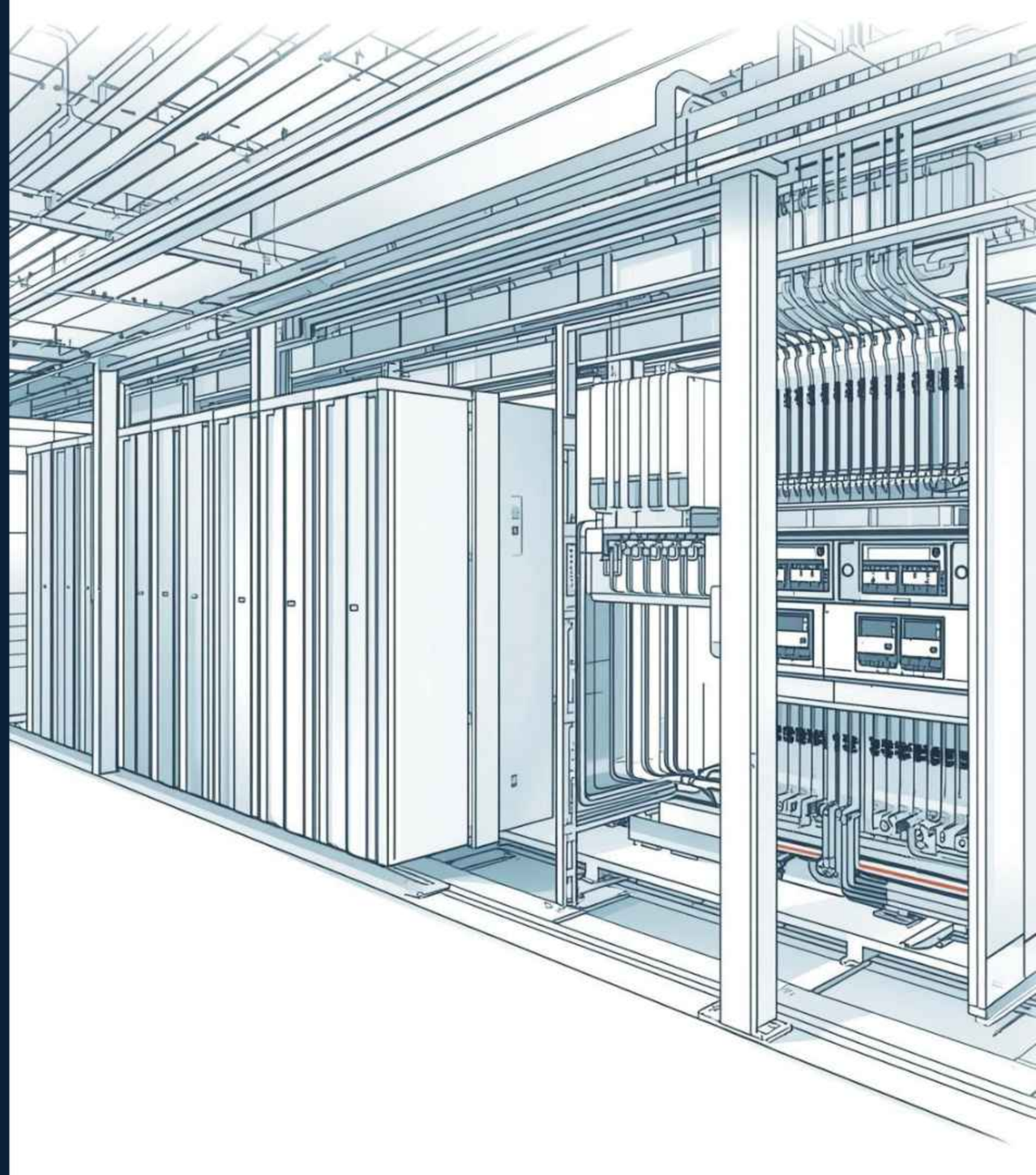


MALGHE

Electricals and Engineers

ELECTRICAL BREAKDOWNS



SYMPTOMS OF ELECTRICAL SYSTEM ISSUES

Motor Failures

Frequent drive failures indicate underlying electrical issues.



Overheating

Cable and switchgear overheating can disrupt operations.



Breaker Tripping

Incorrect breaker tripping can cause significant problems.



MAJOR ELECTRICAL ISSUES FROM CURRENT ABNORMALITIES



Overcurrent

Excessive current can severely damage equipment components.



Short Circuit

Short circuits lead to immediate and catastrophic failures.



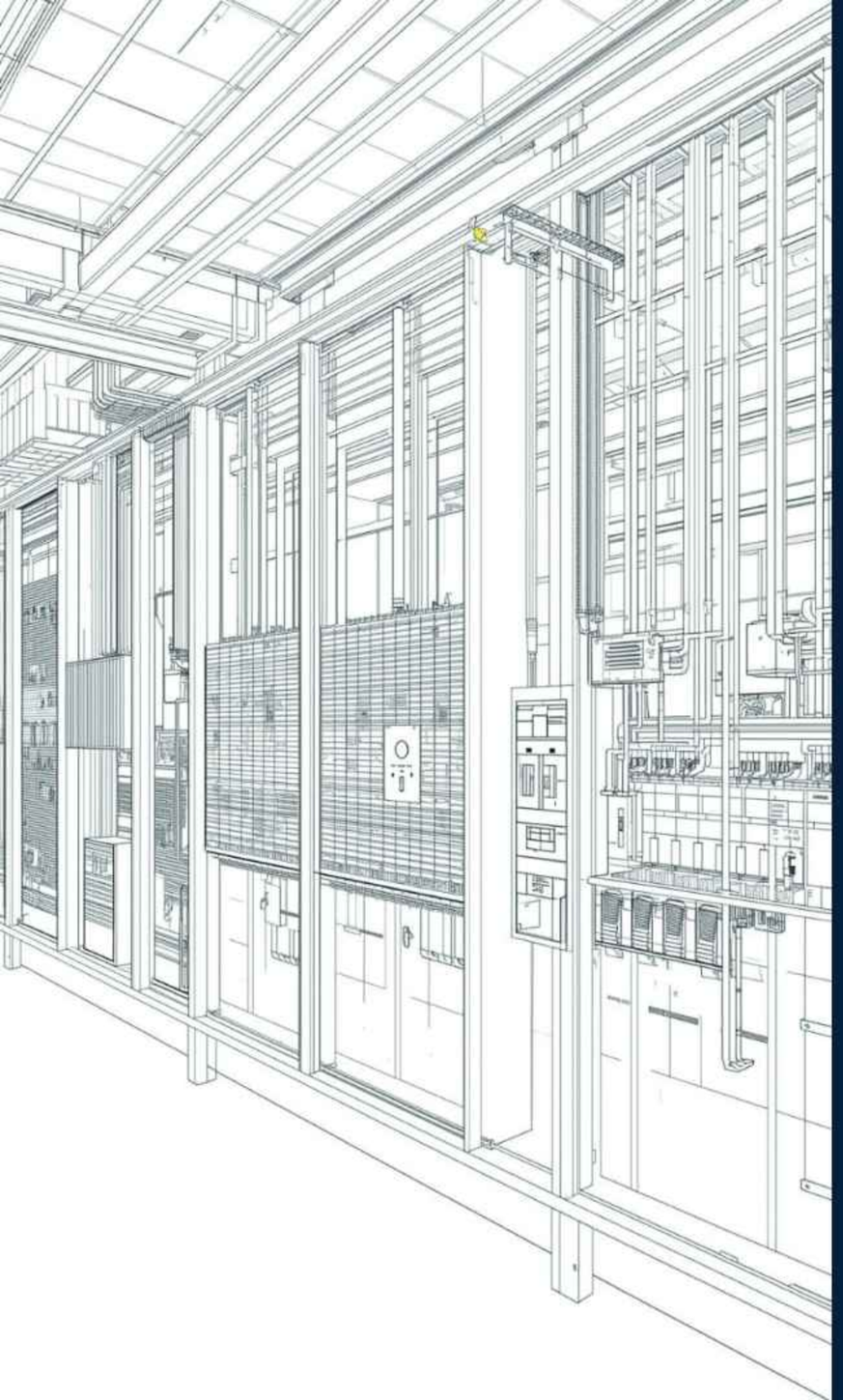
Current Spikes

Sudden spikes put immense stress on system components.



Relay Coordination

Poor coordination causes delays in fault isolation.



DESIGN CAUSES

SELECTION

Incorrect high-tension (HT) and low-tension (LT) switchgear selection leads to insufficient performance and potential equipment failures caused by inappropriate KA ratings.

SIZING

Improper cable sizing significantly increases the risk of overheating, leading to insulation damage, reduced system efficiency, and potential electrical failures.

UNBALANCING

Load unbalancing across phases elevates electrical stress in the system, leading to overheating, reduced equipment efficiency, and potential damage to motors and transformers.

KEY WEAKNESSES IN PROTECTION SYSTEMS

Relay Coordination

Poor relay coordination can lead to system failures.



Incorrect Settings

Incorrect relay settings can compromise safety and operation.



Protection Philosophy

Lack of a cohesive protection philosophy increases risks.

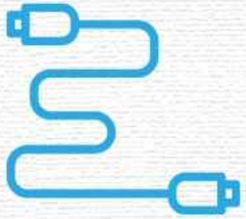


OUR DIAGNOSTIC AND ANALYTICAL APPROACH



System Study

Comprehensive analysis of power system performance.



Cable Verification

Ensuring cables meet operational and safety standards.



CT Validation

Confirming correct selection for protection devices.



Power Analysis

Analyzing power quality and harmonic distortion effects.

EFFECTIVE PROTECTION COORDINATION IN ELECTRICAL SYSTEMS

Tripping Sequence

Ensures orderly disconnection during fault conditions.



Unnecessary Shutdowns

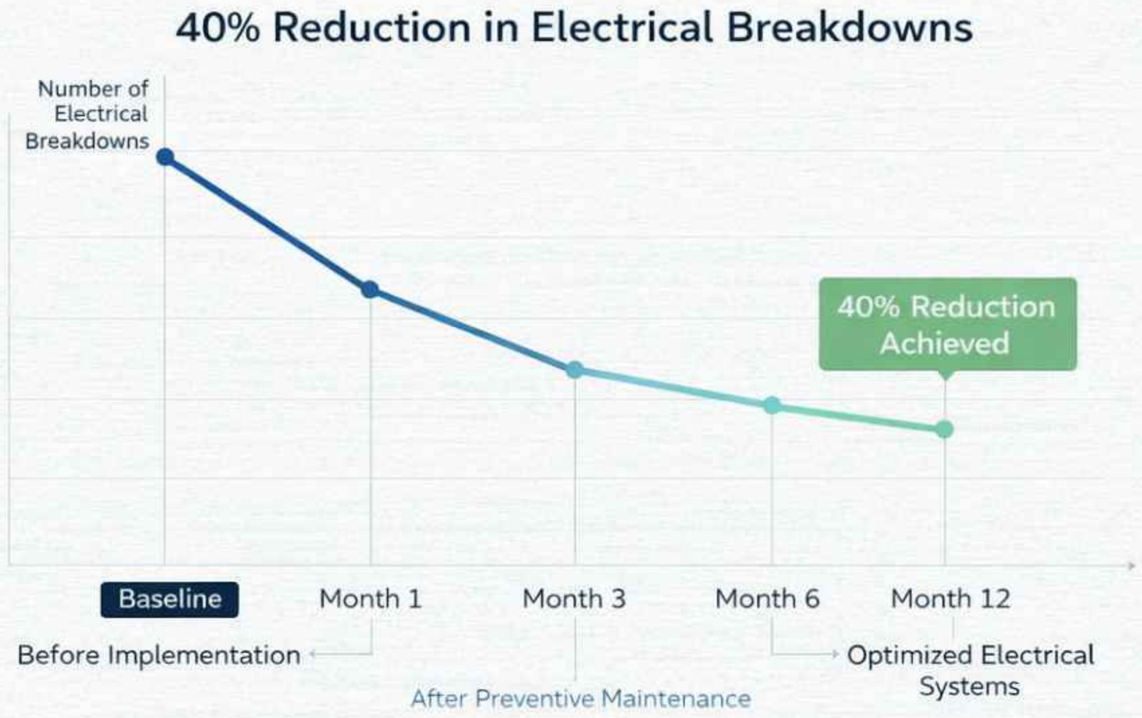
Prevents entire plant shutdowns during minor faults.



Improved Safety

Enhances operational safety and reduces risk exposure.





40% REDUCTION IN ELECTRICAL BREAKDOWNS

By implementing our comprehensive electrical management strategies, industries can significantly reduce electrical breakdowns. This leads to improved system reliability, enhanced operational safety, and lower maintenance costs.

THANK YOU

We appreciate your interest in our solutions
and welcome your questions.

+91 9823580555
marketing@malgheelectricals.com

Malghe Electricals and Engineers Pvt. Ltd.
Pearl Realty, 2nd Floor, Aambethan Chowk,
Chakan, Pune 410 501 India